

Rotations



1 P

2 P

3 D

4 M

5 P

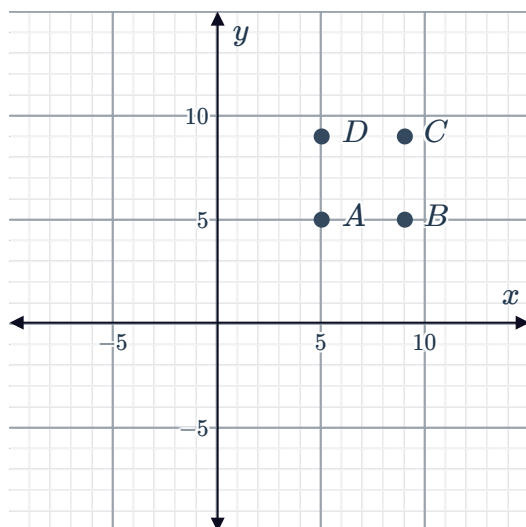
6 G

7 $(0, -2)$

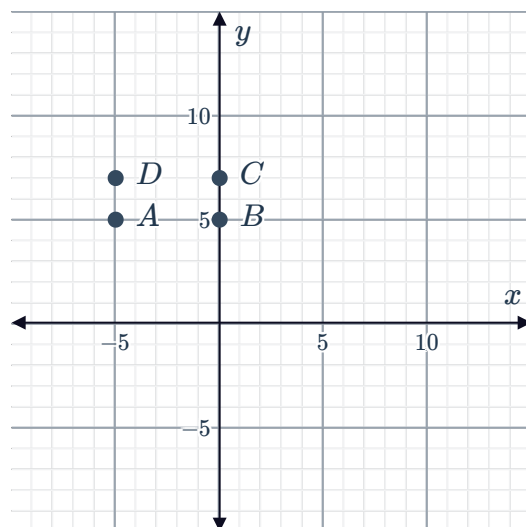
8

a

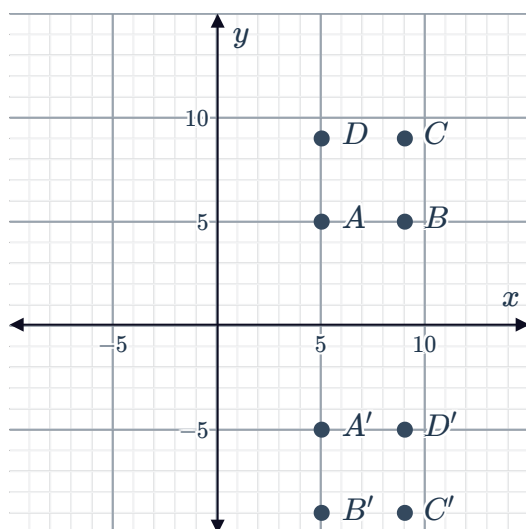
i



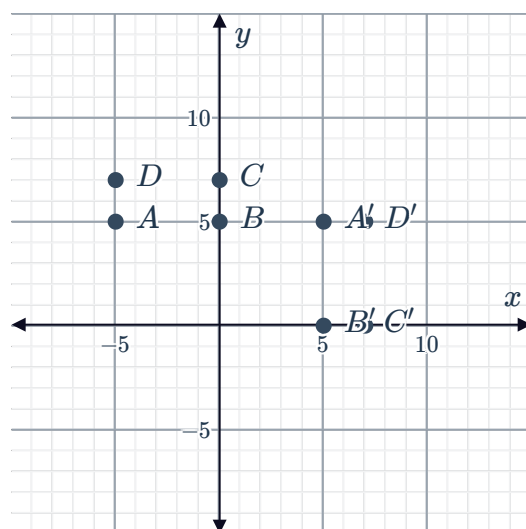
i



ii



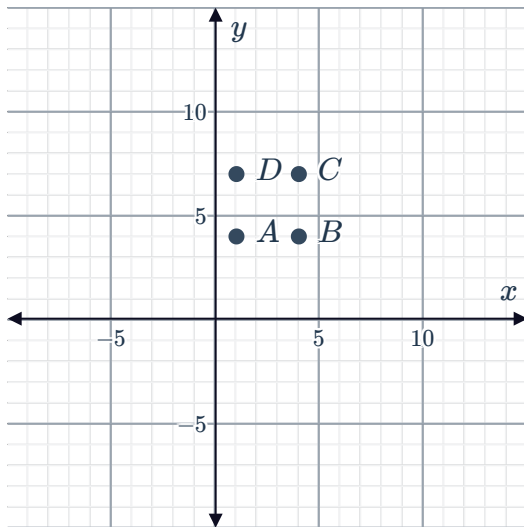
ii



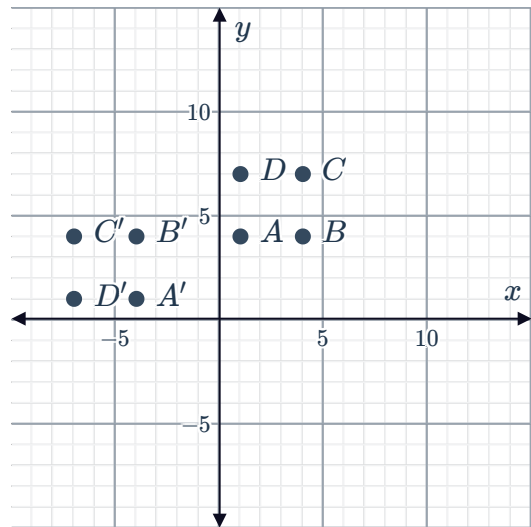
9

a

i

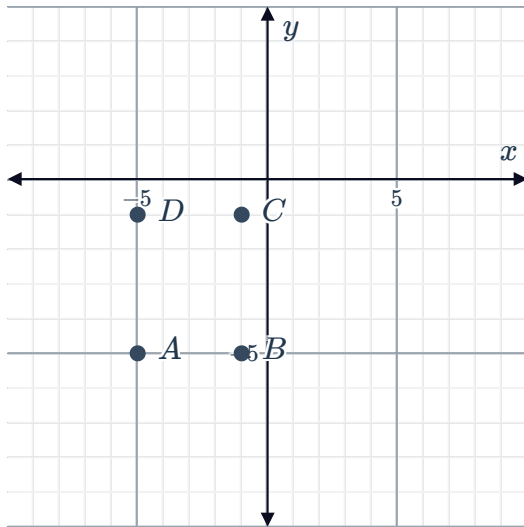


ii

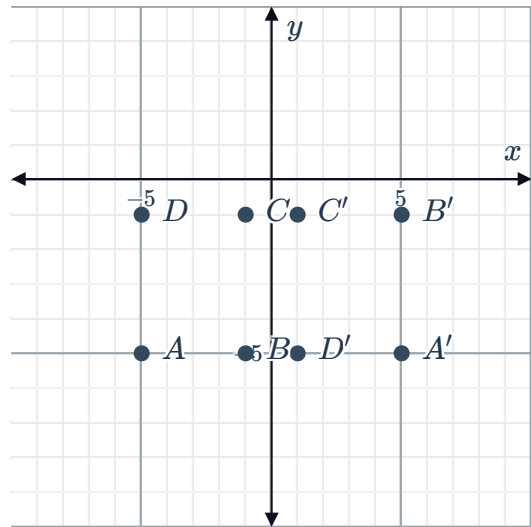


b

i



ii

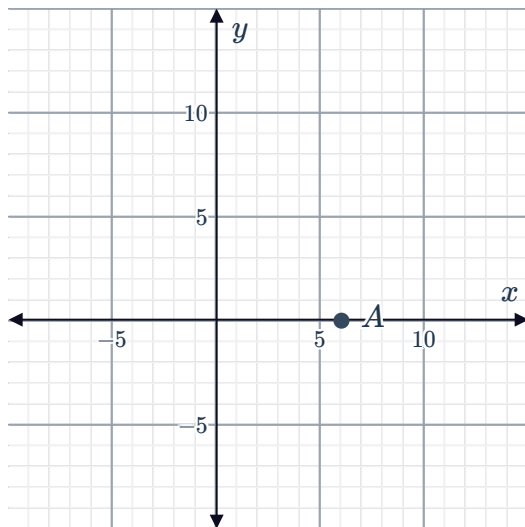


10

a



b

 $B(0, -6)$ 

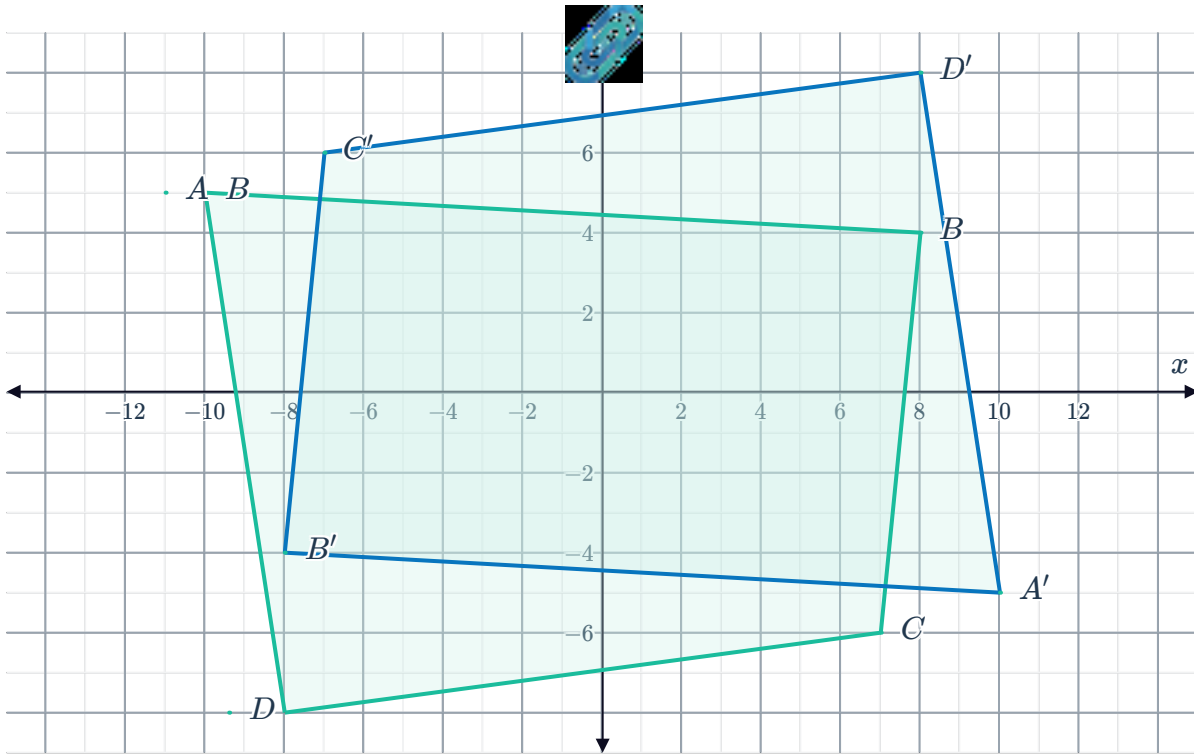
- c 6-unit translation horizontally to the left
6-unit translation vertically downwards

Additional questions

- 11 The shape itself does not change, has been rotated one quadrant clockwise. It is the equivalent of turning the graph on its right side and relabelling the x and y axes.

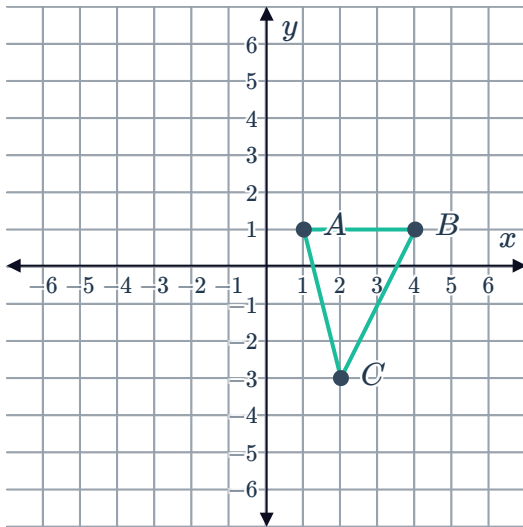
12 $\triangle P$ 13 $(4, -4)$

14

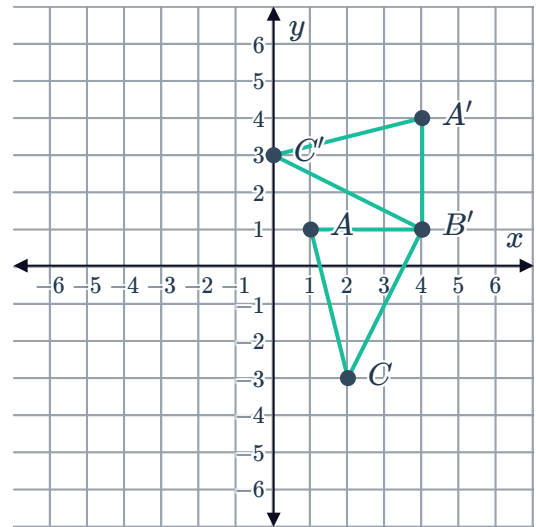


15

a



b



16 Figure A was rotated 90° clockwise about the origin.

17

- a Yes, when rotated 180° each face is showing the dots in the same arrangement as before they were rotated.
- b No, consider the face with one/four/five dot(s). Rotating that 90 gives the exact same face.

18 Avery needs to turn the map 180° clockwise or counterclockwise. He is currently facing



19 Yes, Aneta rotated the segment clockwise around the origin instead of counterclockwise.